

SQL JOINS

```
mysql> CREATE DATABASE school_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE school_db;
Database changed
mysql> CREATE TABLE Student (
  -> student_id INT PRIMARY KEY,
  -> Student_name VARCHAR(100) NOT NULL,
  -> phone_no INT,
  -> depatment_id INT
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Student (student_id, Student_name, phone_no, deparment_id) VALUES
  -> (12, 'Aarav Sharma', 987654321, 1),
  -> (29, 'Diya Patel', 876543210, 2),
  -> (19, 'Vivaan Reddy', 765432109, 3),
  -> (34, 'Ananya Iyer', 654321098, 1),
  -> (54, 'Kabir Singh', 543210987, 4),
  -> (13, 'Ruthu Gowda', 783990987, 2);
Query OK, 6 rows affected (0.02 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE department1 (
  -> department_id INT,
  -> department_name VARCHAR(100) NOT NULL
  -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO department1 (department_id, department_name) VALUES
  -> (1, 'Computer Science'),
  -> (2, 'Electronics Engineering'),
  -> (3, 'Mechanical Engineering'),
  -> (4, 'Information Technology');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
-> FROM Student
-> LEFT JOIN department1 ON Student.depatment_id = department1.department_id;
```

student_id	Student_name	department_name
12	Aarav Sharma	Computer Science
13	Ruthu Gowda	Electronics Engineering
19	Vivaan Reddy	Mechanical Engineering
29	Diya Patel	Electronics Engineering
34	Ananya Iyer	Computer Science
54	Kabir Singh	Information Technology

6 rows in set (0.00 sec)

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
-> FROM Student
-> INNER JOIN department1 ON Student.depatment_id = department1.department_id;
```

student_id	Student_name	department_name
12	Aarav Sharma	Computer Science
13	Ruthu Gowda	Electronics Engineering
19	Vivaan Reddy	Mechanical Engineering
29	Diya Patel	Electronics Engineering
34	Ananya Iyer	Computer Science
54	Kabir Singh	Information Technology

6 rows in set (0.01 sec)

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
-> FROM Student
-> RIGHT JOIN department1 ON Student.depatment_id = department1.department_id;
```

student_id	Student_name	department_name
34	Ananya Iyer	Computer Science
12	Aarav Sharma	Computer Science
29	Diya Patel	Electronics Engineering
13	Ruthu Gowda	Electronics Engineering
19	Vivaan Reddy	Mechanical Engineering
54	Kabir Singh	Information Technology

6 rows in set (0.00 sec)

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
-> FROM Student
-> LEFT JOIN department1 ON Student.depatment_id = department1.department_id
-> UNION
-> SELECT Student.student_id, Student.Student_name, department1.department_name
-> FROM Student
-> RIGHT JOIN department1 ON Student.depatment_id = department1.department_id;
```

student_id	Student_name	department_name
12	Aarav Sharma	Computer Science
13	Ruthu Gowda	Electronics Engineering
19	Vivaan Reddy	Mechanical Engineering
29	Diya Patel	Electronics Engineering
34	Ananya Iyer	Computer Science
54	Kabir Singh	Information Technology

6 rows in set (0.01 sec)